

AI Engineer

Step 3: Learn Deep Learning

CareerCompass — Detailed Learning Notes

What this step is about

Understand neural networks and modern deep-learning workflows.

Core Concepts

- Learn neurons, layers, activation functions, loss, backpropagation, optimizers, batches and epochs.
- Understand training vs validation curves, regularization and learning-rate choices.
- Study dense networks, CNNs and sequence/transformer architectures at a practical level.
- Start with small experiments and inspect results before scaling compute.

Skills You Should Be Able To Demonstrate

- Explain forward pass and backpropagation.
- Train a small neural network.
- Read training/validation curves.
- Compare optimizers.
- Recognize overfitting.

Hands-on Project / Practice

- Train a small image classifier or tabular neural network.
- Run at least two controlled experiments and record the results.

Recommended References

- TensorFlow Documentation — <https://www.tensorflow.org/learn>
- PyTorch Tutorials — <https://pytorch.org/tutorials/>
- Deep Learning Course for Beginners — <https://www.youtube.com/watch?v=HJd1l3FdSnY>